



Team Upsilon 7 - Open Rescue Maze

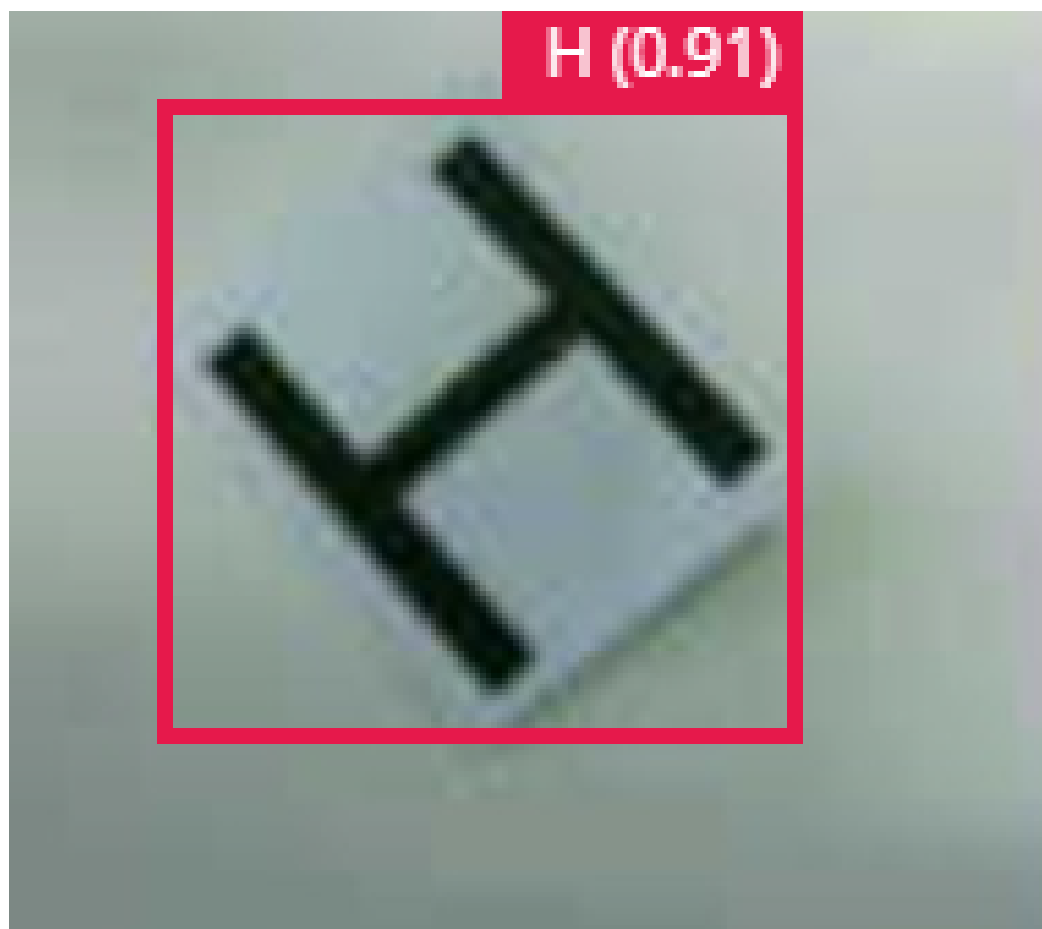


Figure 1 - Victim detection model example identification

Facts

Team Members

Hardware, Electrical & Manufacturing - Ethan Seymour

Software Navigation - James Cousins

Software Victims - Ethan Seymour & Connor Duncan

Success

2026 Robocup Junior Rescue Maze Internationals (South Korea) - Competed

2026 RoboCup Junior Open Rescue Maze Brisbane Regionals - 1st

2025 RoboCup Junior Open Rescue Maze Nationals - 1st

2025 RoboCup Junior Rescue Maze Internationals (Brazil) - Competed

Methods

Challenges Encounter

While designing and testing the robot, some challenges were faced. Most notably when the camera was mounted too far away from the wall for the identification to see the victims, this was caused because the code and hardware was implemented too close to competition and not tested prior.

Lessons Learned

After encountering this challenge our team learned to integrate systems along the way and test all systems together after completing a single system by itself.

Innovations

Our Pivoting LRF system on the front and rear of the robot allows our LRF sensors to remain facing flat at the wall at all times. This prevents the sensors seeing too far or too short because they are being angled as the robot moves over speedbumps or debris. This pivot is being moved using a 180 degree 9g servo motor, with the angle of the robot being determined by the GY-BNO055 6 axis IMU.

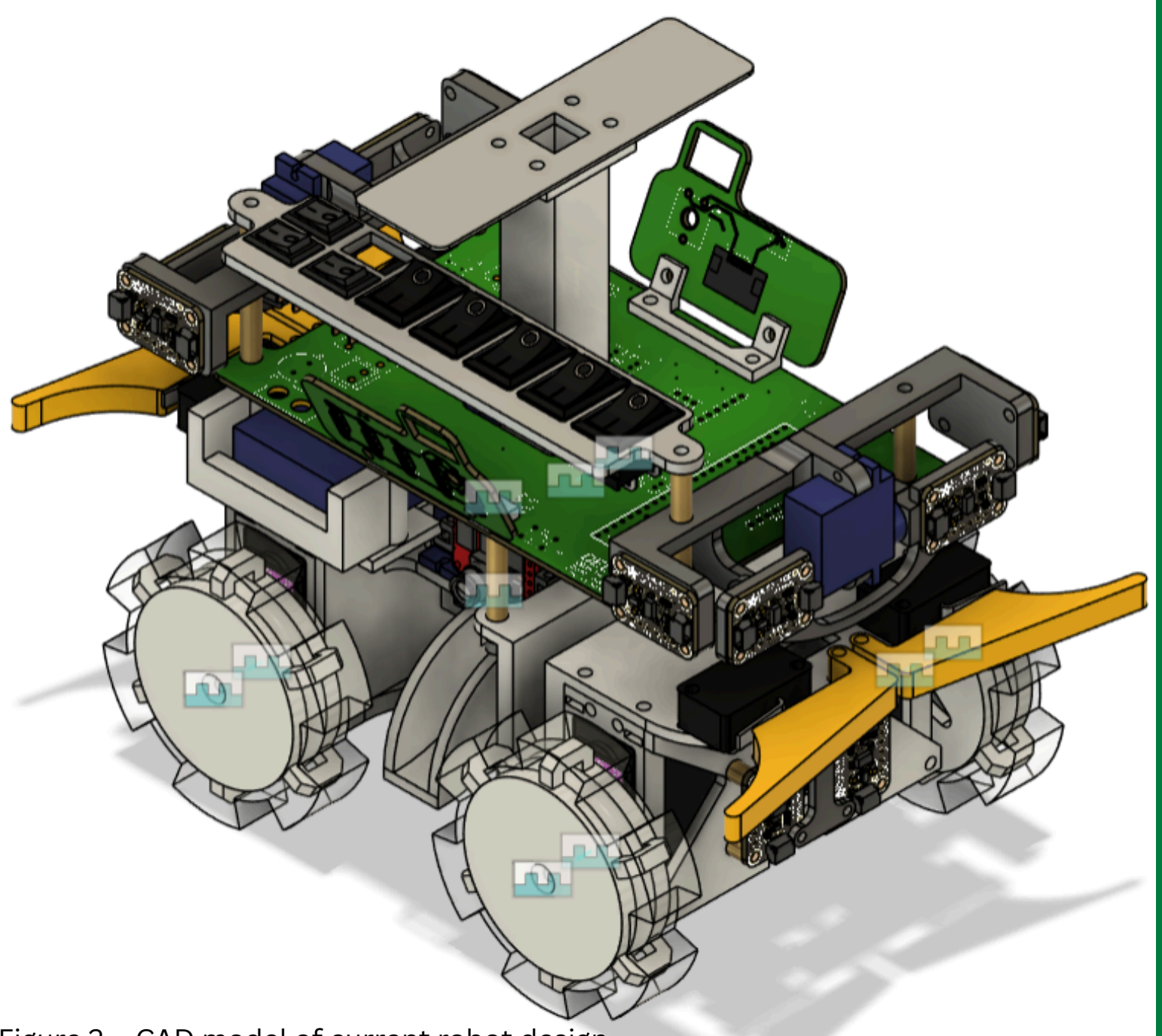


Figure 2 - CAD model of current robot design